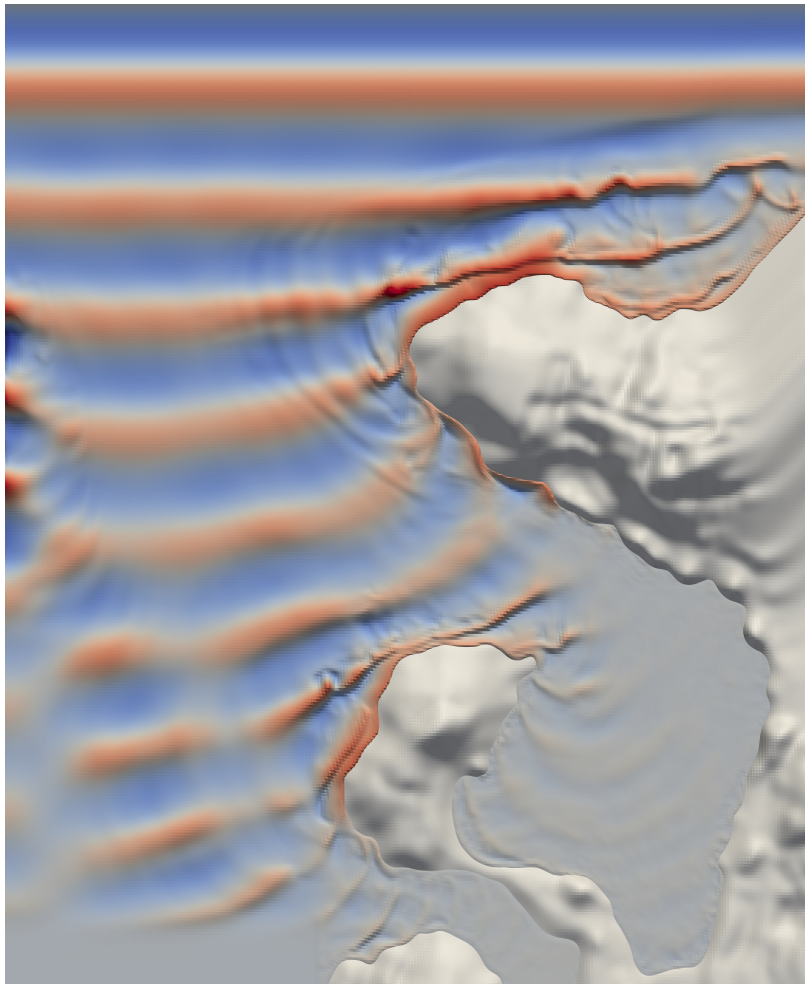


DIVEMesh :: User Guide



DIVEMesh 26.09



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AI declaration

This User Guide has been partially created with the assistance of Claude (Anthropic, model claude-opus-5-5).

Parts of the DIVEMesh open-source software have been created with the assistance of Claude (Anthropic, model claude-opus-5-5).

Contents

1	The ‘control.txt’ file	1
1.1	B :: Boundary	1
1.2	C :: Channel	5
1.3	D :: Data Interpolation	7
1.4	G :: Geodat	9
1.5	H :: Hydrodynamic Coupling	13
1.6	M :: MPI	14
1.7	S :: Solid	15
1.8	T :: Topo	21

Chapter 1

The 'control.txt' file

1.1 B :: Boundary

B 1 `double` cell size dx

default: 0.0

B 2 `int` number of cells in x-, y- and z-direction

When activated, it will overwrite B 1.

default: 0 ; 0 ; 0

B 3 `double` global grid origin delta for the coastal modeling system (CMS) $\Delta x, \Delta y$

default: 0.0 ; 0.0

B 4 `double` global grid rotation for the coastal modeling system (CMS) α

default: 0.0

B 6 `int` turn on coastal modeling system (CMS) (grid rotation and translation)

0 OFF

1 ON

default: 0

B 10 * `double` rectangular domain; $x_{start}, x_{end}, y_{start}, y_{end}, z_{start}, z_{end}$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

B 101 [int](#) type of grid stretching function in x-direction

- 0** OFF
 - 1** center focus (sinh, using B111)
 - 2** wall focus (tanh, using B111)
 - 5** point focus using sinh (requires B 114)
 - 6** point focus using exponents (requires B 114)
 - 8** three zones with constant dx (requires B 121)
 - 9** three zones with stretching (requires B 124)
 - 10** input from file (requires x-spacing.dat file with grid points between 0 and 1)
 - 11** cell size based (requires B 127)
- default:** 0

B 102 [int](#) type of grid stretching function in y-direction

- 0** OFF
 - 1** center focus (sinh, using B112)
 - 2** wall focus (tanh, using B112)
 - 5** point focus using sinh (requires B 115)
 - 6** point focus using exponents (requires B 115)
 - 8** three zones with constant dx (requires B 122)
 - 9** three zones with stretching (requires B 125)
 - 10** input from file (requires y-spacing.dat file with grid points between 0 and 1)
 - 11** cell size based (requires B 128)
- default:** 0

B 103 [int](#) type of grid stretching function in z-direction

- 0** OFF
- 1** center focus (sinh, using B113)
- 2** wall focus (tanh, using B113)
- 3** lid focus (exponential, using B113)
- 4** lid focus (sinusoidal)
- 5** point focus using sinh (requires B 116)
- 6** point focus using exponents (requires B 116)
- 8** three zones with constant dx (requires B 123)
- 9** three zones with stretching (requires B 126)
- 10** input from file (requires z-spacing.dat file with grid points between 0 and 1)

11 cell size based (requires B 129)

default: 0

B 111 **double** stretching factor in x-direction

default: 1.0

B 112 **double** stretching factor in y-direction

default: 1.0

B 113 **double** stretching factor in z-direction

default: 1.0

B 114 **double** focus point for x-direction stretching B 101 5

default: 0.0

B 115 **double** focus point for y-direction stretching B 102 5

default: 0.0

B 116 **double** focus point for z-direction stretching B 103 5

default: 0.0

B 121 input for the three zones of B 101 8: **int** N1, **double** x1, **int** N2, **double** x2, **int** N3

The sum of the given N1,N2,N3 result in the number of elements in x-direction.

default: na

B 122 input for the three zones of B 101 8: **int** N1, **double** y1, **int** N2, **double** y2, **int** N3

The sum of the given N1,N2,N3 result in the number of elements in y-direction.

default: na

B 123 input for the three zones of B 101 8: `int` N1, `double` z1, `int` N2, `double` z2, `int` N3

The sum of the given N1,N2,N3 result in the number of elements in z-direction.

default: na

B 124 input for the three zones of B 101 9: `int` N1, `double` x1, `double` f1, `int` N2, `double` xf, `double` f2, `int` N3, `double` x2, `double` f3

The sum of the given N1,N2,N3 result in the number of elements in x-direction. The factors f1 and f3 are the linear stretching factors for the layers. The parameter x1 gives the border between the first and second layer, x2 between second and third. The parameter xf gives the location of the stretching focus in the second layer, where sinh stretching based on the factor f2 is used.

default: na

B 125 input for the three zones of B 102 9: `int` N1, `double` y1, `double` f1, `int` N2, `double` yf, `double` f2, `int` N3, `double` y2, `double` f3

see B 124.

default: na

B 126 input for the three zones of B 103 9: `int` N1, `double` z1, `double` f1, `int` N2, `double` zf, `double` f2, `int` N3, `double` z2, `double` f3

see B 124.

default: na

B 127 input for the cell size based stretching of B 101 11: `double` Δx_{min} , `double` Δx_{max} , `double` x_f , `double` δ_f , `double` r

A constant cell size of Δx_{min} is generated in the focus zone of length δ_f around x_f . The cells are stretched with the cell ratio r between the ends of the focus zone and the domain boundaries until the maximum cell size of Δx_{max} is reached.

default: na

B 128 input for the cell size based stretching of B 102 11: `double` Δy_{min} , `double` Δy_{max} ,
`double` y_f , `double` δ_f , `double` r

see B 127

default: na

B 129 input for the cell size based stretching of B 103 11: `double` Δz_{min} , `double` Δz_{max} ,
`double` z_f , `double` δ_f , `double` r

see B 127

default: na

B 130 `int` print grid to "gridSpacing.vtk"

0 OFF

1 ON

default: 0

1.2 C :: Channel

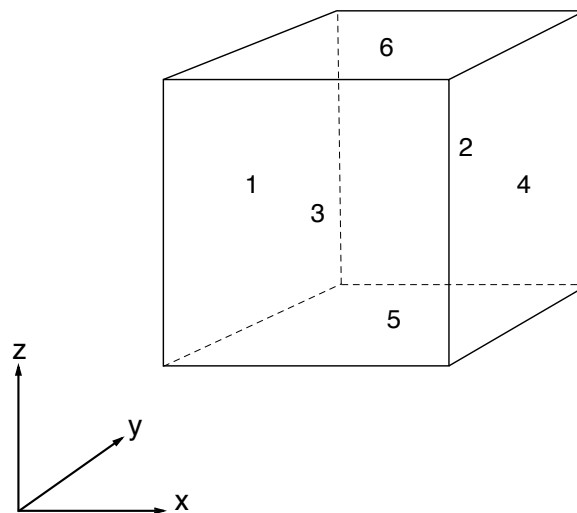


Figure 1.1: Definition of cell sides within DIVEMesh and REEF3D.

C 11 `int` Boundary Condition on Surfside 1

1 inflow

3 symmetry plane

6 wave generation

7 numerical beach

21 wall

default: 21

C 12 [int](#) Boundary Condition on Surfside 2

For 2D simulations, this is set to symmetry plane automatically.

3 symmetry plane

6 wave generation

7 numerical beach

21 wall

default: 21

C 13 [int](#) Boundary Condition on Surfside 3

For 2D simulations, this is set to symmetry plane automatically.

3 symmetry plane

6 wave generation

7 numerical beach

21 wall

default: 21

C 14 [int](#) Boundary Condition on Surfside 4

2 outflow

3 symmetry plane

6 wave generation

7 numerical beach

21 wall

default: 21

C 15 [int](#) Boundary Condition on Surfside 5

3 symmetry plane

21 wall

default: 21

C 16 `int` Boundary Condition on Surfside 6

3 symmetry plane

21 wall

default: 21

C 21 `int` Periodic boundary conditions in x-direction

0 OFF

1 ON

default: 0

C 22 `int` Periodic boundary conditions in y-direction

0 OFF

1 ON

default: 0

C 23 `int` Periodic boundary conditions in z-direction

0 OFF

1 ON

default: 0

1.3 D :: Data Interpolation

D 10 `int` turn data interpolation on/off

0 OFF

1 ON

default: 0

D 11 `double` Δx , Δy , Δz

default: 0.0 ; 0.0 ; 0.0

D 12 `double` factor x-coordinate, factor y-coordinate, factor z-coordinate,

default: 0.0 ; 0.0 ; 0.0

D 13 `int` Read data every i^{th} iteration

default: 1

D 14 `int` Type of interpolation

1 inverse distance

2 kriging

default: 1

D 15 `int` Number of smoothing iterations

default: 0

D 16 `double` Factor for data smoothing

default: 0.5

D 17 `double` Factor for data inverse distance

default: 16.0

D 18 `double` Range factor for kriging

default: 0.3

D 19 `int` Read dummy letter in front of 3D coordinates

0 OFF

1 ON

default: 0

D 23 `int` reverse sign of data variable in dmdata.dat

0 OFF

1 ON

default: 0

D 24 `double` distance limiter for inverse distance

default: na

1.4 G :: Geodat

G 9 **int** geodata for topo or solid

1 topo

2 solid

default: 1

G 10 **int** turn geodat on/off

0 OFF

1 ON

default: 0

G 11 **double** Δx , Δy , Δz

default: 0.0 ; 0.0 ; 0.0

G 12 **double** factor x-coordinate, factor y-coordinate, factor z-coordinate,

default: 0.0 ; 0.0 ; 0.0

G 13 **double** rotation angle of geo coordinates around vertical axis

default: 0.0

G 14 **double** x-coordinate and y-coordinate of origin for the rotation angle of geo coordinates around vertical axis

default: 0.0 ; 0.0

G 15 **int** interpolation scheme

Global inverse distance interpolation requires long computation times for large geodat sets. The recommended interpolation scheme for relatively homogeneously spaced geodat points is local inverse distance interpolation. Kriging is recommended for inhomogeneously spaced geodat points, e.g. geodat points from digitized map contour lines. Kriging becomes unstable when geodat points are spaced too close to each other, so for this option G 36 is turned on automatically with the default value of 1.0 (that value can be changed). Kriging becomes inefficient for geodat sets larger than 10,000 points, thus G 37 is recommended for kriging.

1 global inverse distance interpolation

2 local inverse distance interpolation

3 kriging

4 local kriging

default: 2

G 17 **int** d_{ij} lower limit for local inverse distance interpolation

default: 2

G 18 **int** number of required points for local inverse distance interpolation

default: 14

G 19 **int** read a letter in front of the coordinates in the geo.dat file

0 OFF

1 ON

default: 0

G 20 **int** use automatic grid size

0 OFF

1 ON

default: 0

G 21 **double** Margins for automatic grid size $x_{start}, y_{start}, z_{start}$

default: 0.0 ; 0.0 ; 0.0

G 22 **double** Margins for automatic grid size $x_{end}, y_{end}, z_{end}$

default: 0.0 ; 0.0 ; 0.0

G 23 **int** reverse sign of vertical coordinate in geo.dat file

0 OFF

1 ON

default: 0

G 24 **double** raise topography above the level h by dz

default: 0.0 ; 0.0

G 25 **double** multiply topography above the level h by factor fz

default: 0.0 ; 0.0

G 26 **double** remove all geodat points below the bed level h

default: 0.0

G 27 **double** remove all geodat points above the bed level h

default: 0.0

G 31 **int** Number of outer spatial filter iterations

default: 0

G 32 **int** Number of inner spatial filter iterations

default: 1

G 35 **double** Factor for data inverse distance

default: 16.0

G 36 **double** sampling geodat points: factor times horizontal average mesh size for duplicate geodat point identification

default: 1.0

G 37 **int** sampling geodat points: maximum number of geodat after random removal

default: 1e8

G 38 **int** skip horizontal cells for geodat interpolation algorithm

default: 1

G 39 **int** remove out-of-bounds geodata points

0 OFF

1 ON

default: 1

G 41 **int** print SWAN bottom file from interpolated geo points

0 OFF

1 ON

default: 0

G 51 **int** automatic holecheck and holefill for incomplete geodata sets

G 52 is used for elevation for areas without points.

0 OFF

1 ON

default: 0

G 52 **double** base topography value for local inverse distance interpolation

default: 0.0

G 53 **double** automatic search radius factor (times dx)

default: 0.0

G 60 **int** read "geo_patch.dat" file to patch/augment the "geo.dat" file.

0 OFF

1 ON

default: 0

G 61 **int** add the geo_patch points to the geo points

0 OFF

1 ON

default: 0

G 62 **double** factor for geo_patch kernel radius (factor time horizontal mesh size)

default: 3.0

1.5 H :: Hydrodynamic Coupling

H 10 **int** Turn on hydrodynamic coupling procedure

This option requires the presence of the corresponding state files and state file folders.

0 OFF

2 SFLOW to CFD

4 FNPF to CFD

5 NHFLOW to CFD

44 FNPF to FNPF

default: 0

H 21 **double** X-location in FNPF NWT which is origin in CFD NWT

default: 0.0

H 22 **double** Y-location in FNPF NWT which is origin in CFD NWT

default: 0.0

H 23 **double** Z-location in FNPF NWT which is origin in CFD NWT

default: 0.0

H 31 **double** start time coupling time series

default: -10.0^{19}

H 32 **double** end time coupling time series

default: 10.0^{19}

H 33 **int** start iteration coupling time series

default: -21^8

H 34 `int` end iteration coupling time series

default: 21^8

1.6 M :: MPI

M 10 `int` Number of processes

default: 1

M 11 `int` Partition in x-direction

0 OFF

1 ON

default: 1

M 12 `int` Partition in y-direction

0 OFF

1 ON

default: 1

M 13 `int` Partition in z-direction

0 OFF

1 ON

default: 0

M 20 `int` Decomposition method

1 Standard rectangular base domain

2 Improved rectangular base domain

default: 1

M 31 `double` Variance parameter for decomposition method 2

default: 0.1

1.7 S :: Solid

S 1 `int` read STL file and generate solid. Name the file as 'solid.stl'

IMPORTANT: the STL file needs to be in ASCII format! Most CAD programs export to a binary STL file. It is possible to use e.g. Paraview for STL conversion from binary to ASCII.

0 OFF

1 ON

default: 0

S 2 `int` auto-generate mesh extend based on STL max/min coordinates

0 OFF

1 ON

default: 0

S 3 `double` margins for auto-generated mesh $x_{m_{start}}, x_{m_{end}}, y_{m_{start}}, y_{m_{end}}, z_{m_{start}}, z_{m_{end}}$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

S 4 `double` Scale STL geometry

default: 1.0

S 5 `double` 3D rotation; $x_{origin}, y_{origin}, z_{origin}, \phi, \theta, \psi$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

S 6 `int` print transformed STL model to "REEF3D_Solid.stl"

0 OFF

1 ON

default: 0

S 7 `double` translation / change origin of STL model dx, dy, dz

default: 0.0 ; 0.0 ; 0.0

S 8 `double` Turn STL geometry in horizontal xy-plane in degree $^{\circ}$

default: 0.0

S 9 **int** Invert inside/outside for STL geometry

1 regular

2 invert

default: 1

S 10 * **double** rectangular object; $x_{start}, x_{end}, y_{start}, y_{end}, z_{start}, z_{end}$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

S 11 * **double** rectangular object array; $x_{origin}, y_{origin}, z_{origin}$, box length L, gap G, number of objects in each direction n_i, n_j, n_k

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

S 12 * **double** beam with rectangular profile with flexible orientation; $x_{start}, y_{start}, z_{start}, x_{end}, y_{end}, z_{end}$, b, h

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

S 15 **int** rotate STL geometry together with the geo.dat points

0 OFF

1 ON

default: 0

S 32 * **double** cylinder in y-direction; $x_{center}, z_{center}, radius$

default: 0.0 ; 0.0 ; 0.0

S 33 * **double** cylinder in z-direction; $x_{center}, y_{center}, radius$

default: 0.0 ; 0.0 ; 0.0

S 37 * **double** cylinder with flexible orientation and front face orthogonal to the cylinder axis

$x_{start}, y_{start}, z_{start}, radius_{start}, x_{end}, y_{end}, z_{end}, radius_{end}$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

S 41 * **double** cone in x-direction; $y_{center}, z_{center}, x_{start}, x_{end}, radius_{start}, radius_{end}$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

S 42 * **double** cone in y-direction; $x_{center}, z_{center}, y_{start}, y_{end}, radius_{start}, radius_{end}$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

S 43 * **double** cone in z-direction; $x_{center}, y_{center}, z_{start}, z_{end}, radius_{start}, radius_{end}$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

S 51 * **double** sphere; $x_{center}, y_{center}, z_{center}, radius$

default: 0.0 ; 0.0 ; 0.0 ; 0.0

S 52 * **double** ellipsoid; $x_{center}, y_{center}, z_{center}, a_{axis}, b_{axis}, c_{axis}$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

S 53 * **double** semi ellipsoid with vertical base; $x_{center}, y_{center}, z_{center}, a_{axis}, b_{axis}, c_{axis}, h_{base}$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

S 54 * **double** semi ellipsoid with vertical base with rotation around the center;

$x_{center}, y_{center}, z_{center}, a_{axis}, b_{axis}, c_{axis}, h_{base}, \phi, \theta, \psi$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

S 61 * **double** wedge object in x-direction; $x_{start}, x_{end}, y_{start}, y_{end}, z_{start}, z_{end}$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

S 62 * **double** wedge object in y-direction; $x_{start}, x_{end}, y_{start}, y_{end}, z_{start}, z_{end}$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

S 63 * **double** wedge object in z-direction; $x_{start}, x_{end}, y_{start}, y_{end}, z_{start}, z_{end}$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

S 81 * **double** tetrahedon object, each of the 4 points is given by the coordinates $x_1, y_1, z_1, x_2, y_2, z_2, x_3, y_3, z_3, x_4, y_4, z_4$

default: [4x] 0.0 ; 0.0 ; 0.0

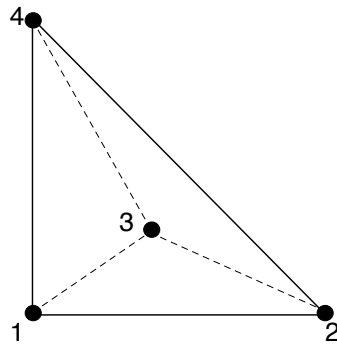


Figure 1.2: Definition of the tetrahedon points.

S 82 * **double** pyramid object, each of the 5 points is given by the coordinates $x_1, y_1, z_1, x_2, y_2, z_2, x_3, y_3, z_3, x_4, y_4, z_4, x_5, y_5, z_5$

default: [5x] 0.0 ; 0.0 ; 0.0

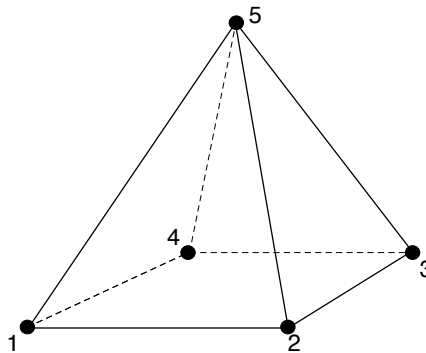


Figure 1.3: Definition of the pyramid points.

S 83 * **double** wedge object, each of the 6 points is given by the coordinates $x_1, y_1, z_1, x_2, y_2, z_2, x_3, y_3, z_3, x_4, y_4, z_4, x_5, y_5, z_5, x_6, y_6, z_6$

default: [6x] 0.0 ; 0.0 ; 0.0

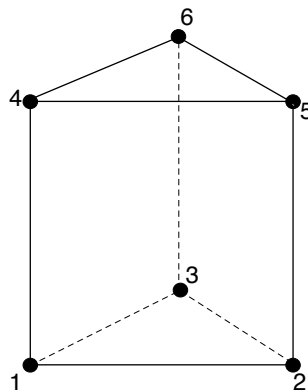


Figure 1.4: Definition of the wedge points.

S 84 * **double** hexahedron object, each of the 8 points is given by the coordinates $x_1, y_1, z_1, x_2, y_2, z_2, x_3, y_3, z_3, x_4, y_4, z_4, x_5, y_5, z_5, x_6, y_6, z_6, x_7, y_7, z_7, x_8, y_8, z_8$

default: [8x] 0.0 ; 0.0 ; 0.0

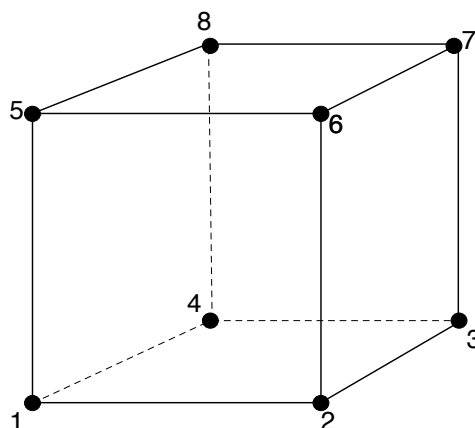


Figure 1.5: Definition of the hexahedron points.

S 121 * **double** vertical ogee weir, coordinates of upstream bottom corner, width, downstream height and hydraulic head x, y, z, b, P_d, H_0

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

S 122 * **double** vertical ogee weir, K, n, x_c and y_c

default: 0.5 ; 1.85 ; 0.22 ; 0.075

S 123 * **double** vertical ogee weir, R_1 and R_2

default: 0.45 ; 0.2

S 131 * **double** semi-circular cylinder in y-direction, x_c , z_c , y_{start} , y_{end} , $radius_{start}$, $radius_{end}$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

S 141 * **double** arch bridge; x_{start} , x_{end} , y_{start} , y_{end} , z_{start} , z_{end} , $radius$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

S 301 **int** fluvial box geometry.

The inverted fluvial box geometry only works for special cases, where the inflow is on side 1 and the outflow is on side 4.

1 regular

2 inverted

default: 1

S 305 **double** fluvial box resolution, factor time dx results in ds (smaller values gives higher resolution); *factor*

default: 0.5

S 306 **double** fluvial box width; b

default: 1.0

S 307 **double** fluvial box flow heighth; fh , bed height bh

default: 0.5 ; 0.2

S 308 **double** fluvial box move origin; Δx , Δy , Δz ,

default: 0.0 ; 0.0 ; 0.0

S 309 **double** fluvial box margins; Δx , Δy , Δz ,

default: 0.0 ; 0.0 ; 0.0

S 310 * **double** fluvial box: straight segment; *length*

default: 0.0

S 320 * **double** fluvial box: left bend; *radius, angle ϕ*

default: 0.0

S 330 * **double** fluvial box: right bend; *radius, angle ϕ*

default: 0.0

S 340 * **double** fluvial box: meander (sine-generated curve); *θ, L, N, ds*

default: 0.0 ; 0.0 ; 0.0 ; 0.0

1.8 T :: Topo

T 10 * **double** rectangular object; *$x_{start}, x_{end}, y_{start}, y_{end}, z_{start}, z_{end}$*

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

T 11 * **double** rectangular object array; *$x_{origin}, y_{origin}, z_{origin}$, box length L, gap G, number of objects in each direction n_i, n_j, n_k*

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

T 32 * **double** cylinder in y-direction; *$x_{center}, z_{center}, radius$*

default: 0.0 ; 0.0 ; 0.0

T 33 * **double** cylinder in z-direction; *$x_{center}, y_{center}, radius$*

default: 0.0 ; 0.0 ; 0.0

T 37 * **double** cylinder with flexible orientation and front face orthogonal to the cylinder axis

$x_{start}, y_{start}, z_{start}, radius_{start}, x_{end}, y_{end}, z_{end}, radius_{end}$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

T 41 * **double** cone in x-direction; *$y_{center}, z_{center}, x_{start}, x_{end}, radius_{start}, radius_{end}$*

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

T 42 * **double** cone in y-direction; $x_{center}, z_{center}, y_{start}, y_{end}, radius_{start}, radius_{end}$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

T 43 * **double** cone in z-direction; $x_{center}, y_{center}, z_{start}, z_{end}, radius_{start}, radius_{end}$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

T 51 * **double** sphere; $x_{center}, y_{center}, z_{center}, radius$

default: 0.0 ; 0.0 ; 0.0 ; 0.0

T 52 * **double** ellipsoid; $x_{center}, y_{center}, z_{center}, a_{axis}, b_{axis}, c_{axis}$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

T 53 * **double** semi ellipsoid with vertical base; $x_{center}, y_{center}, z_{center}, a_{axis}, b_{axis}, c_{axis}, h_{base}$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

T 54 * **double** semi ellipsoid with vertical base with rotation around the center;

$x_{center}, y_{center}, z_{center}, a_{axis}, b_{axis}, c_{axis}, h_{base}, \phi, \theta, \psi$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

T 61 * **double** wedge object in x-direction; $x_{start}, x_{end}, y_{start}, y_{end}, z_{start}, z_{end}$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

T 62 * **double** wedge object in y-direction; $x_{start}, x_{end}, y_{start}, y_{end}, z_{start}, z_{end}$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

T 63 * **double** wedge object in z-direction; $x_{start}, x_{end}, y_{start}, y_{end}, z_{start}, z_{end}$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

T 81 * **double** tetrahedon object, each of the 4 points is given by the coordinates $x_1, y_1, z_1, x_2, y_2, z_2, x_3, y_3, z_3, x_4, y_4, z_4$

default: [4x] 0.0 ; 0.0 ; 0.0

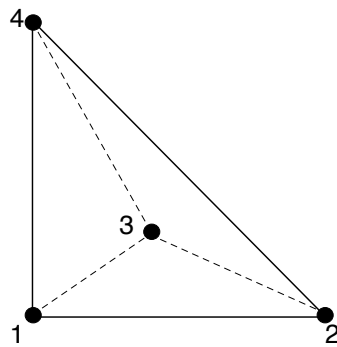


Figure 1.6: Definition of the tetrahedon points.

T 82 * **double** pyramid object, each of the 5 points is given by the coordinates $x_1, y_1, z_1, x_2, y_2, z_2, x_3, y_3, z_3, x_4, y_4, z_4, x_5, y_5, z_5$

default: [5x] 0.0 ; 0.0 ; 0.0

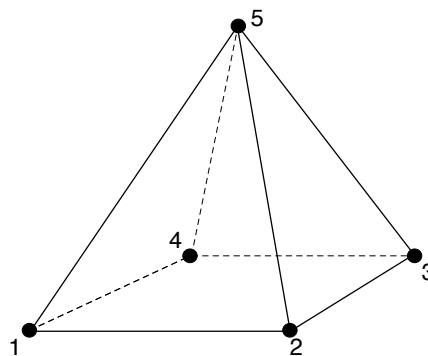


Figure 1.7: Definition of the pyramid points.

T 83 * **double** wedge object, each of the 6 points is given by the coordinates $x_1, y_1, z_1, x_2, y_2, z_2, x_3, y_3, z_3, x_4, y_4, z_4, x_5, y_5, z_5, x_6, y_6, z_6$

default: [6x] 0.0 ; 0.0 ; 0.0

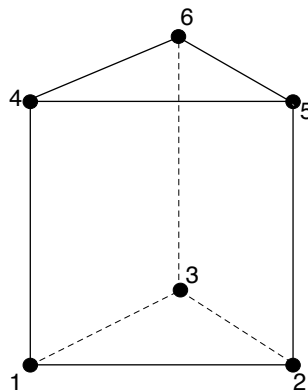


Figure 1.8: Definition of the wedge points.

T 84 * **double** hexahedron object, each of the 8 points is given by the coordinates $x_1, y_1, z_1, x_2, y_2, z_2, x_3, y_3, z_3, x_4, y_4, z_4, x_5, y_5, z_5, x_6, y_6, z_6, x_7, y_7, z_7, x_8, y_8, z_8$

default: [8x] 0.0 ; 0.0 ; 0.0

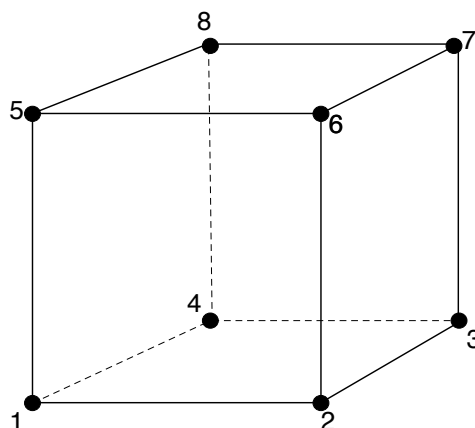


Figure 1.9: Definition of the hexahedron points.

T 121 * **double** vertical ogee weir, coordinates of upstream bottom corner, width, downstream height and hydraulic head x, y, z, b, P_d, H_0

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

T 122 * **double** vertical ogee weir, K, n, x_c and y_c

default: 0.5 ; 1.85 ; 0.22 ; 0.075

T 123 * **double** vertical ogee weir, R_1 and R_2

default: 0.45 ; 0.2

T 131 * **double** semi-circular cylinder in y-direction, x_c , z_c , y_{start} , y_{end} , $radius_{start}$, $radius_{end}$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

T 141 * **double** arch bridge; x_{start} , x_{end} , y_{start} , y_{end} , z_{start} , z_{end} , $radius$

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

T 301 **int** fluvial box geometry.

The inverted fluvial box geometry only works for special cases, where the inflow is on side 1 and the outflow is on side 4.

1 regular

2 inverted

default: 1

T 305 **double** fluvial box resolution, factor time dx results in ds (smaller values gives higher resolution); *factor*

default: 0.5

T 306 **double** fluvial box width; b

default: 1.0

T 307 **double** fluvial box flow heighth; fh , bed height bh

default: 0.5 ; 0.2

T 308 **double** fluvial box move origin; Δx , Δy , Δz ,

default: 0.0 ; 0.0 ; 0.0

T 309 **double** fluvial box margins; Δx , Δy , Δz ,

default: 0.0 ; 0.0 ; 0.0

T 310 * **double** fluvial box: straight segment; *length*

default: 0.0

T 320 * **double** fluvial box: left bend; *radius, angle ϕ*

default: 0.0

T 330 * **double** fluvial box: right bend; *radius, angle ϕ*

default: 0.0

T 340 * **double** fluvial box: meander (sine-generated curve); *θ, L, N, ds*

default: 0.0 ; 0.0 ; 0.0 ; 0.0